

Beyond the Triangle

Trigonometry as a Mathematical Bridge

UGB Mastery Classes | April 2026

Objective

Transition from structural identities to cross-disciplinary problem-solving: how trigonometry quietly threads through pure geometry, complex numbers, polynomial algebra, and combinatorics.

Each problem in this session is solved by translating it into a different language — complex exponentials, convexity, polynomial roots, geometric ratios — and reading the answer off in the original.

Problem 1 — Statement

Problem 1

Let $\sin^5 \theta = a \sin 5\theta + b \sin 3\theta + c \sin \theta$. Find a, b, c .

Direct expansion of $\sin^5 \theta$ in elementary form is opaque. The route is to write $\sin \theta$ as a difference of complex exponentials, expand by the binomial theorem, and collect the resulting powers of $z = e^{i\theta}$ into multiple-angle sines.

Problem 1 — Proof, Part 1

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Let $z = e^{i\theta}$, so that $\sin \theta = \frac{z - z^{-1}}{2i}$. By the binomial theorem,

$$\sin^5 \theta = \left(\frac{z - z^{-1}}{2i} \right)^5 = \frac{1}{32i^5} (z - z^{-1})^5 = \frac{1}{32i} (z - z^{-1})^5.$$

Expanding $(z - z^{-1})^5$,

$$(z - z^{-1})^5 = z^5 - 5z^3 + 10z - 10z^{-1} + 5z^{-3} - z^{-5}.$$

Group symmetric pairs:

$$(z - z^{-1})^5 = (z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1}).$$

Problem 1 — Proof, Part 2

Problem 1 — Proof, Part 2

Using $z^k - z^{-k} = 2i \sin(k\theta)$,

$$\sin^5 \theta = \frac{1}{32i} \left[2i \sin 5\theta - 5 \cdot 2i \sin 3\theta + 10 \cdot 2i \sin \theta \right].$$

The i 's cancel and we obtain

$$\sin^5 \theta = \frac{1}{16} \sin 5\theta - \frac{5}{16} \sin 3\theta + \frac{5}{8} \sin \theta.$$

Reading off the coefficients,

$$a = \frac{1}{16}, \quad b = -\frac{5}{16}, \quad c = \frac{5}{8}.$$

Generalization for Odd Powers

The same procedure works for every odd power $2n + 1$. Writing $\sin \theta = (z - z^{-1}) / (2i)$ and applying the binomial theorem,

$$\sin^{2n+1} \theta = \frac{1}{4^n} \sum_{k=0}^n (-1)^{n-k} \binom{2n+1}{k} \sin[(2n+1-2k)\theta].$$

As an example, for $n = 4$ (i.e. $\sin^9 \theta$),

$$\sin^9 \theta = \frac{1}{2^8} \left[\binom{9}{0} \sin 9\theta - \binom{9}{1} \sin 7\theta + \binom{9}{2} \sin 5\theta - \dots \right].$$

Quick Check 1 — Even Power of Cosine

Question

Let $\cos^6 \theta = a \cos 6\theta + b \cos 4\theta + c \cos 2\theta + d$. Which option lists (a, b, c, d) correctly?

- (A) $\left(\frac{1}{32}, \frac{3}{16}, \frac{15}{32}, \frac{5}{16}\right)$
- (B) $\left(\frac{1}{16}, \frac{3}{16}, \frac{15}{16}, \frac{5}{16}\right)$
- (C) $\left(\frac{1}{32}, \frac{6}{32}, \frac{15}{32}, \frac{20}{32}\right)$
- (D) $\left(\frac{1}{64}, \frac{6}{64}, \frac{15}{64}, \frac{20}{64}\right)$

Hint: write $\cos \theta = (z + z^{-1})/2$ and expand $(z + z^{-1})^6$ using the binomial theorem; pair z^k with z^{-k} to get $2 \cos(k\theta)$.

Quick Check 1 — Answer

Quick Check 1 — Answer

Answer: (A).

Using $\cos \theta = (z + z^{-1})/2$,

$$\cos^6 \theta = \frac{1}{64} (z + z^{-1})^6 = \frac{1}{64} \sum_{k=0}^6 \binom{6}{k} z^{6-2k}.$$

Pairing terms $z^j + z^{-j} = 2 \cos(j\theta)$ and noting $\binom{6}{0} = \binom{6}{6} = 1$, $\binom{6}{1} = \binom{6}{5} = 6$, $\binom{6}{2} = \binom{6}{4} = 15$, $\binom{6}{3} = 20$,

$$\begin{aligned} \cos^6 \theta &= \frac{1}{64} [2 \cos 6\theta + 12 \cos 4\theta + 30 \cos 2\theta + 20] \\ &= \frac{1}{32} \cos 6\theta + \frac{3}{16} \cos 4\theta + \frac{15}{32} \cos 2\theta + \frac{5}{16}. \end{aligned}$$

Hence $(a, b, c, d) = (\frac{1}{32}, \frac{3}{16}, \frac{15}{32}, \frac{5}{16})$. The other options either drop a power of 2 in the denominator or fail to merge the $\binom{6}{k}$ pairs.

Convexity and Midpoints

Theoretical Foundation

A function f is strictly concave on an interval if, for every pair $x_1 < x_2$,

$$f\left(\frac{x_1 + x_2}{2}\right) > \frac{f(x_1) + f(x_2)}{2}.$$

Jensen's inequality extends this to n variables: the value of f at the arithmetic mean strictly exceeds the arithmetic mean of the function values, provided f is concave.

Problem 2 — Statement

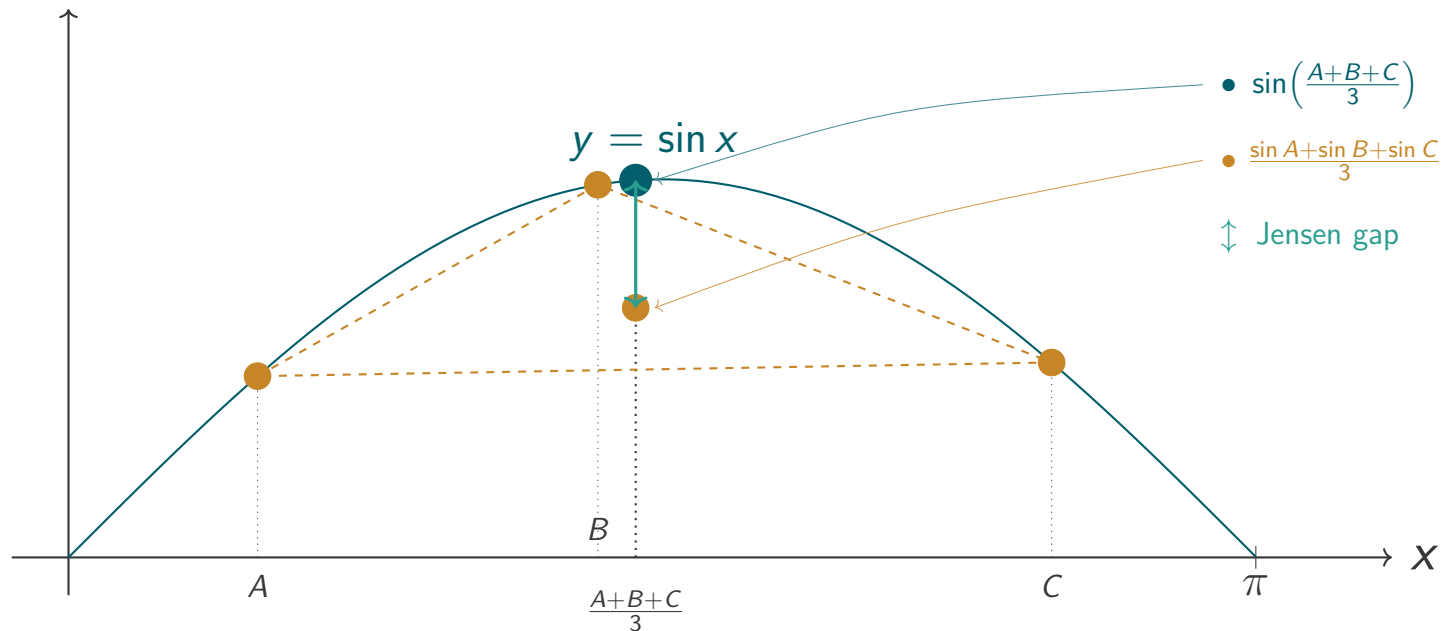
Problem 2

Let A, B, C be the angles of a triangle. Prove that

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}.$$

The constraint $A + B + C = \pi$ is symmetric in the three angles, and the function $\sin$ is concave on $(0, \pi)$. Jensen's inequality, applied to a concave function under a fixed-sum constraint, immediately produces the bound.

Geometric Picture: Jensen ($n = 3$)



The dashed triangle joins three points on the sine curve. Its centroid (the average of the three function values) sits below the curve at the average of the three angles — that vertical gap is exactly Jensen's inequality.

Problem 2 — Proof

Problem 2 — Proof

Consider $f(x) = \sin x$ on $(0, \pi)$. Since $f''(x) = -\sin x < 0$ throughout this interval, f is strictly concave.

By Jensen's inequality applied to $A, B, C \in (0, \pi)$,

$$\frac{\sin A + \sin B + \sin C}{3} \leq \sin\left(\frac{A + B + C}{3}\right).$$

Since $A + B + C = \pi$, the right-hand side equals $\sin(\pi/3) = \sqrt{3}/2$. Multiplying by 3,

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}.$$

Equality holds iff $A = B = C = \pi/3$, i.e. for the equilateral triangle. □

Trigonometry and Circle Radii

Theoretical Foundation

A powerful dictionary translates trigonometric sums into the metric invariants of a triangle's circumradius R and inradius r :

$$\cos A + \cos B + \cos C = 1 + \frac{r}{R}.$$

This identity bridges the algebraic properties of angles with the geometric constraints imposed by the inscribed and circumscribed circles. Establishing it is the heart of the proof of Euler's inequality $R \geq 2r$.

Problem 3 — Statement

Problem 3

Prove Euler's geometric inequality $R \geq 2r$, by first establishing the trigonometric identity

$$\cos A + \cos B + \cos C = 1 + \frac{r}{R}.$$

The plan is in two stages: collapse the cosine sum into the product form $1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$, then identify this product with r/R via the standard formula for the inradius.

Problem 3 — Proof, Part 1

Problem 3 — Proof, Part 1

Apply sum-to-product to the first two cosines:

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}.$$

For the third, the double-angle identity gives $\cos C = 1 - 2 \sin^2 \frac{C}{2}$.

Since $A + B + C = \pi$, we have $\cos \frac{A+B}{2} = \sin \frac{C}{2}$, so the full sum factors as

$$\cos A + \cos B + \cos C = 1 + 2 \sin \frac{C}{2} \left[\cos \frac{A-B}{2} - \sin \frac{C}{2} \right].$$

Replacing $\sin \frac{C}{2}$ inside the bracket with $\cos \frac{A+B}{2}$ and applying sum-to-product once more,

$$\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}.$$

Problem 3 — Proof, Part 2

Problem 3 — Proof, Part 2

The standard geometric relation for the inradius is

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2},$$

so the bracketed expression equals r/R , establishing the identity

$$\cos A + \cos B + \cos C = 1 + \frac{r}{R}.$$

By concavity of cosine (Jensen, as in Problem 2),

$$\cos A + \cos B + \cos C \leq 3 \cos \frac{\pi}{3} = \frac{3}{2}.$$

Substituting the identity,

$$1 + \frac{r}{R} \leq \frac{3}{2} \implies \boxed{R \geq 2r.} \quad \square$$

Problem 4 — Statement

Problem 4

For any integer $n \geq 2$, evaluate the finite product

$$\prod_{k=1}^{n-1} \sin \frac{k\pi}{n}.$$

The key observation is that the angles $k\pi/n$ are exactly half the arguments of the non-trivial n -th roots of unity. The product of $1 - \zeta_k$ over those roots is computable from the polynomial $z^{n-1} + \cdots + 1$, and modulus considerations isolate the sines.

Problem 4 — Proof, Part 1

Problem 4 — Proof, Part 1

The n -th roots of unity are $z_k = e^{i2k\pi/n}$ for $k = 0, 1, \dots, n-1$, satisfying $z^n - 1 = 0$. Factoring out the trivial root $z = 1$,

$$z^{n-1} + z^{n-2} + \dots + 1 = \prod_{k=1}^{n-1} (z - z_k).$$

Evaluating both sides at $z = 1$,

$$n = \prod_{k=1}^{n-1} (1 - e^{i2k\pi/n}).$$

Each factor admits a half-angle factorisation:

$$1 - e^{i2\theta} = e^{i\theta}(e^{-i\theta} - e^{i\theta}) = e^{i\theta} \cdot (-2i \sin \theta).$$

Problem 4 — Proof, Part 2

Problem 4 — Proof, Part 2

Setting $\theta = k\pi/n$,

$$n = \prod_{k=1}^{n-1} \left(-2i e^{ik\pi/n} \sin \frac{k\pi}{n} \right).$$

Take the modulus of both sides. Each $e^{ik\pi/n}$ has modulus 1, $|-2i| = 2$, and $\sin(k\pi/n) > 0$ for $1 \leq k \leq n-1$, so

$$n = 2^{n-1} \prod_{k=1}^{n-1} \sin \frac{k\pi}{n} \implies \boxed{\prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = \frac{n}{2^{n-1}}.} \quad \square$$

The Tangent Bijection

Theoretical Foundation

The tangent function gives a bijection $(-\frac{\pi}{2}, \frac{\pi}{2}) \xrightarrow{\sim} \mathbb{R}$. The algebraic constraint $x + y + z = xyz$ mirrors the identity

$$\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - (\tan A \tan B + \tan B \tan C + \tan C \tan A)}.$$

Setting the numerator to zero matches the constraint exactly, forcing $A + B + C = \pi$ (modulo π). When $x, y, z > 0$, the angles fall in $(0, \pi/2)$ and form an acute triangle.

Problem 5 — Statement

Problem 5

Given positive reals x, y, z satisfying $x + y + z = xyz$, prove that

$$\frac{1}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+y^2}} + \frac{1}{\sqrt{1+z^2}} \leq \frac{3}{2}.$$

The expression $1/\sqrt{1+x^2}$ shouts trigonometric substitution: setting $x = \tan A$ converts each radical into $\cos A$. The algebraic constraint then enforces $A + B + C = \pi$, reducing the problem to a cosine-sum inequality already in our toolkit.

Problem 5 — Proof

Problem 5 — Proof

Set $x = \tan A$, $y = \tan B$, $z = \tan C$ with $A, B, C \in (0, \pi/2)$. The constraint $x + y + z = xyz$ forces $A + B + C = \pi$ — in particular, A, B, C are the angles of an acute triangle.

Each radical simplifies via the Pythagorean identity:

$$\frac{1}{\sqrt{1 + \tan^2 A}} = \frac{1}{\sec A} = \cos A.$$

The inequality therefore reduces to

$$\cos A + \cos B + \cos C \leq \frac{3}{2},$$

which we already established (by Jensen, or as part of Problem 3). □

Problem 6 — Statement

Problem 6

Find a closed-form expression for the combinatorial sum

$$\sum_{k=0}^n \binom{n}{k} \cos(k\theta).$$

Problem 6 — Proof, Part 1

Problem 6 — Proof, Part 1

Embed the real sum inside a complex one:

$$C + iS = \sum_{k=0}^n \binom{n}{k} [\cos(k\theta) + i \sin(k\theta)] = \sum_{k=0}^n \binom{n}{k} (e^{i\theta})^k.$$

By the binomial theorem this collapses to $(1 + e^{i\theta})^n$.

Factor a half-angle exponential from the base:

$$1 + e^{i\theta} = e^{i\theta/2} (e^{-i\theta/2} + e^{i\theta/2}) = e^{i\theta/2} \cdot 2 \cos \frac{\theta}{2}.$$

Raising to the n -th power,

$$(1 + e^{i\theta})^n = e^{in\theta/2} (2 \cos \frac{\theta}{2})^n.$$

Problem 6 — Proof, Part 2

Problem 6 — Proof, Part 2

Expanding the exponential into real and imaginary parts,

$$(1 + e^{i\theta})^n = (2 \cos \frac{\theta}{2})^n [\cos \frac{n\theta}{2} + i \sin \frac{n\theta}{2}].$$

Extracting the real component,

$$\sum_{k=0}^n \binom{n}{k} \cos(k\theta) = 2^n \cos^n \frac{\theta}{2} \cos \frac{n\theta}{2}.$$

As a sanity check at $\theta = 0$: the right-hand side gives $2^n \cdot 1 \cdot 1 = 2^n$, which matches $\sum_k \binom{n}{k} = 2^n$. □

Problem 7 — Statement

Problem 7

For any positive integer m , evaluate the finite sum

$$\sum_{k=1}^m \cot^2 \frac{k\pi}{2m+1}.$$

Problem 7 — Proof, Part 1

Problem 7 — Proof, Part 1

Expand $(\cos \theta + i \sin \theta)^{2m+1}$ via the binomial theorem. The imaginary part equals $\sin((2m+1)\theta)$ and consists of terms where i appears to an odd power:

$$\sin((2m+1)\theta) = \binom{2m+1}{1} \cos^{2m}\theta \sin \theta - \binom{2m+1}{3} \cos^{2m-2}\theta \sin^3\theta + \dots$$

Setting $\sin((2m+1)\theta) = 0$ identifies roots at $\theta = k\pi/(2m+1)$. For $\sin \theta \neq 0$, divide throughout by $\sin^{2m+1}\theta$ to obtain a polynomial of degree m in $x = \cot^2 \theta$:

$$\binom{2m+1}{1} x^m - \binom{2m+1}{3} x^{m-1} + \dots = 0.$$

Problem 7 — Proof, Part 2

Problem 7 — Proof, Part 2

The roots of this polynomial are precisely $x_k = \cot^2 \frac{k\pi}{2m+1}$ for $k = 1, 2, \dots, m$. By Viète's formulas, the sum of roots equals the ratio of the second coefficient to the leading one (with a sign from the alternating pattern $+, -, +, \dots$):

$$\sum_{k=1}^m x_k = \frac{\binom{2m+1}{3}}{\binom{2m+1}{1}} = \frac{(2m+1)(2m)(2m-1)/6}{2m+1} = \frac{m(2m-1)}{3}.$$

Hence

$$\boxed{\sum_{k=1}^m \cot^2 \frac{k\pi}{2m+1} = \frac{m(2m-1)}{3}}. \quad \square$$

Problem 8 — Statement

Problem 8

Using the result of Problem 7 and geometric bounding, prove that

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

Problem 8 — Proof, Part 1

Problem 8 — Proof, Part 1

For $x \in (0, \pi/2)$, the sector–triangle comparison on the unit circle yields $\sin x < x < \tan x$. Squaring and inverting,

$$\cot^2 x < \frac{1}{x^2} < 1 + \cot^2 x.$$

Substitute $x = k\pi / (2m + 1)$ and sum over $k = 1, \dots, m$:

$$\sum_{k=1}^m \cot^2 \frac{k\pi}{2m+1} < \sum_{k=1}^m \frac{(2m+1)^2}{k^2 \pi^2} < m + \sum_{k=1}^m \cot^2 \frac{k\pi}{2m+1}.$$

Injecting the closed form from Problem 7,

$$\frac{m(2m-1)}{3} < \frac{(2m+1)^2}{\pi^2} \sum_{k=1}^m \frac{1}{k^2} < m + \frac{m(2m-1)}{3}.$$

Problem 8 — Proof, Part 2

Problem 8 — Proof, Part 2

Multiply through by $\frac{\pi^2}{(2m+1)^2}$ to isolate the partial sum:

$$\frac{\pi^2}{(2m+1)^2} \cdot \frac{m(2m-1)}{3} < \sum_{k=1}^m \frac{1}{k^2} < \frac{\pi^2}{(2m+1)^2} \left[m + \frac{m(2m-1)}{3} \right].$$

As $m \rightarrow \infty$, both bounds are dominated by their leading terms; each behaves like

$$\frac{2m^2}{4m^2} \cdot \frac{\pi^2}{3} = \frac{\pi^2}{6}.$$

By the squeeze theorem,

$$\boxed{\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}. \quad \square}$$

Problem 9 — Statement

Problem 9

Prove that the roots of the cubic

$$x^3 - 3x + 1 = 0$$

are exactly $2 \cos 40^\circ$, $2 \cos 80^\circ$, and $2 \cos 160^\circ$.

Problem 9 — Proof, Part 1

Problem 9 — Proof, Part 1

Substitute $x = 2 \cos \theta$ into the cubic:

$$(2 \cos \theta)^3 - 3(2 \cos \theta) + 1 = 0 \implies 8 \cos^3 \theta - 6 \cos \theta + 1 = 0.$$

Dividing by 2 and invoking the triple-angle identity $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$,

$$2 \cos 3\theta + 1 = 0 \implies \cos 3\theta = -\frac{1}{2}.$$

Problem 9 — Proof, Part 2

Problem 9 — Proof, Part 2

Over one full period, $\cos 3\theta = -\frac{1}{2}$ has solutions

$$3\theta = 120^\circ, 240^\circ, 480^\circ,$$

yielding the base angles

$$\theta = 40^\circ, 80^\circ, 160^\circ.$$

These produce three distinct values of $2 \cos \theta$, so the roots of $x^3 - 3x + 1 = 0$ are

$$2 \cos 40^\circ, \quad 2 \cos 80^\circ, \quad 2 \cos 160^\circ.$$



Thank you

Questions?